Section A

1(a) power, watt
potential difference, volt
magnetic flux, weber
magnetic flux density, tesla

(b) Speed is a quantity, so it should be defined in terms of other quantities. But "second" is a unit (or second isn't a quantity), so the statement is incorrect.
Speed is the distance travelled per unit time.

(c)(i) horizontal component of the velocity, $v_x = v\cos\theta$
Range, $d = (v\cos\theta)t$

(ii) vertical component of the velocity, $v_y = v\sin\theta$
$s_y = u_y t + (1/2)at^2$
$0 = (v\sin\theta)t + (1/2)(-g)t^2$
$$t = \frac{2v\sin\theta}{g}$$

(iii) $$\sin 2\theta = \frac{dg}{v^2} = \frac{(94)(9.81)}{(32)^2}$$
$= 0.9005$
$2\theta = 64.23^\circ$ or 115.8°
$\theta_1 = 32.1^\circ = 32^\circ$
$\theta_2 = 57.9^\circ = 58^\circ$

2(a)(i) The student's argument is not correct because the reaction force acts on the man